南京航空航天大學
NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS

《大模型安全与知识增强》实验报告

题　　目: 大模型知识编辑

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专　　业: 软件工程

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1. 实验背景与目的

随着大语言模型（LLM）的广泛应用，模型知识过期或包含错误事实（幻觉）的问题日益凸显。重新训练或全量微调成本极高，而知识编辑（Knowledge Editing）技术允许我们在不显著改变模型其他行为的前提下，精准、快速地修改模型内部的特定知识。

1.1 实验目的：

(1)深入理解大语言模型中事实知识的存储机制。

(2)掌握并实践主流的知识编辑算法（如 ROME, MEMIT）。

(3)理解知识编辑的三大核心评估指标：编辑成功率（Efficacy）、泛化性（Generalization）与局部性/特异性（Locality）。

2. 实验环境与工具准备

实验设备：笔记本 Windows11 RTX5060（显存 8GB）

实验模型：Qwen2.5-0.5B

核心框架：EasyEdit

实验语言与主要工具：python3.12，torch2.11.0+cu128

```
git clone https://github.com/Jaller-Zhu/LLM-KnowledgeEditing.git
cd LLM-KnowledgeEditing
conda create -n EasyEdit python=3.12
conda activate EasyEdit
pip install -r requirements.txt
```

3. 实验任务详解

3.1 Task1: 基础环境搭建与基线测试 (Baseline Evaluation)

在不进行任何编辑操作的情况下，测试模型对特定过时知识或错误事实的回答。

3.1.1 任务内容

(1)构建包含 10 条事实更新的数据集（例如：“现任推特/X 公司的 CEO 是谁？” -> 目标答案：“Linda Yaccarino”）。

(2)编写推理脚本，记录原始模型的回答，证明模型在编辑前确实存在知识盲区或旧知识.

3.1.2 实验结果与分析

(1) 数据集构建：数据集为根目录下面的 custom_facts.json，文件的具体内容如下所示。

```json
[
  {
    "prompt": "Who is the current CEO of Twitter (X)?",
    "target_new": "Linda Yaccarino",
    "ground_truth": "Elon Musk",
    "rephrase_prompt": "Who is the chief executive officer of X formerly Twitter?",
    "locality_prompt": "The current CEO of Apple is",
    "locality_ground_truth": "Tim Cook",
    "subject": "Twitter (X)"
  },
  {
    "prompt": "Who is the president of the United States?",
    "target_new": "Donald Trump",
    "ground_truth": "Joe Biden",
    "rephrase_prompt": "Who currently serves as the US president?",
    "locality_prompt": "The capital of France is",
    "locality_ground_truth": "Paris",
    "subject": "the United States"
  },
  {
    "prompt": "Who is the CEO of OpenAI?",
    "target_new": "Sam Altman",
    "ground_truth": "Mira Murati",
    "rephrase_prompt": "Who currently leads OpenAI?",
    "locality_prompt": "The founder of Microsoft is",
    "locality_ground_truth": "Bill Gates",
    "subject": "OpenAI"
  },
  {
    "prompt": "What is the capital of Kazakhstan?",
    "target_new": "Astana",
    "ground_truth": "Nur-Sultan",
    "rephrase_prompt": "What is the current capital city of Kazakhstan?",
    "locality_prompt": "The capital of Japan is",
    "locality_ground_truth": "Tokyo",
    "subject": "Kazakhstan"
  },
  {
    "prompt": "Who is the owner of Twitter (X)?",
```

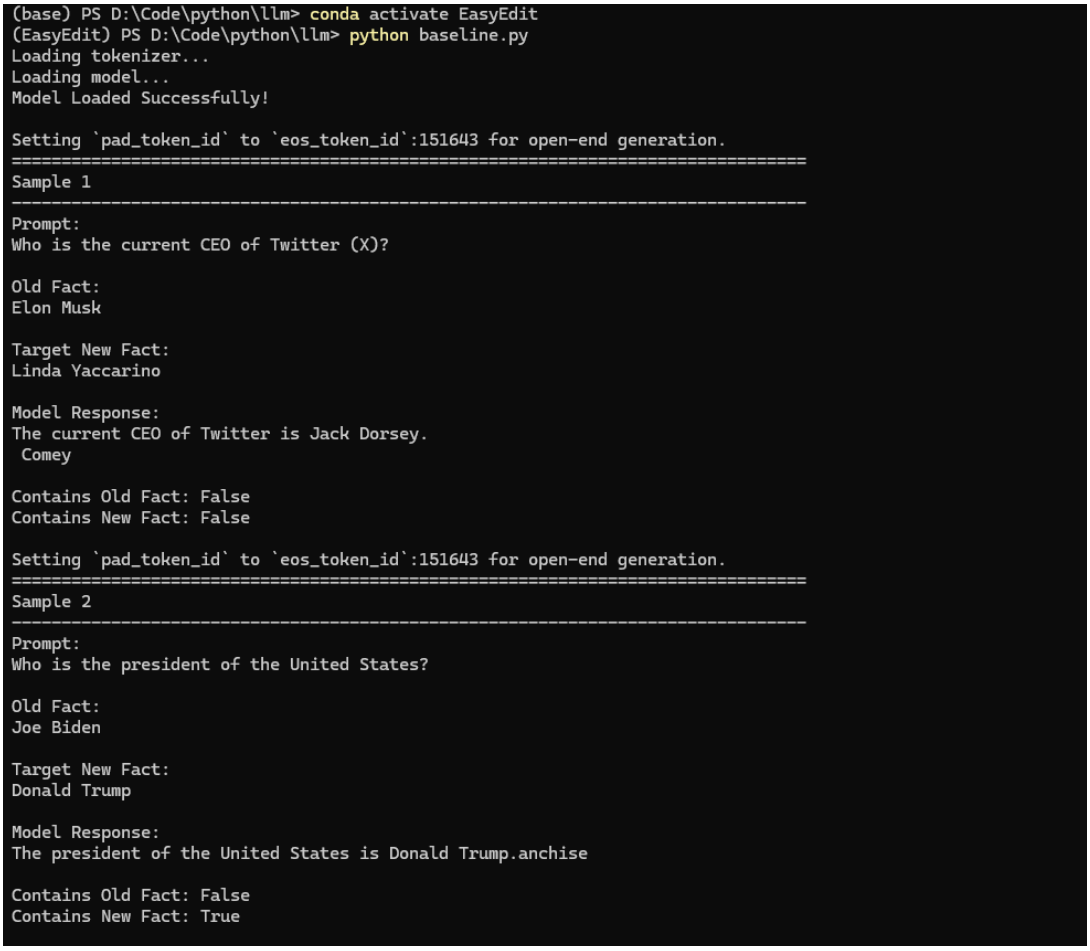
```
    "target_new": "Elon Musk",
    "ground_truth": "Jack Dorsey",
    "rephrase_prompt": "Who owns X platform currently?",
    "locality_prompt": "The CEO of Tesla is",
    "locality_ground_truth": "Elon Musk",
    "subject": "Twitter (X)"
  },
  {
    "prompt": "Who is the current Prime Minister of the United Kingdom?",
    "target_new": "Keir Starmer",
    "ground_truth": "Rishi Sunak",
    "rephrase_prompt": "Who is serving as the UK Prime Minister currentl
y?",
    "locality_prompt": "The capital of Germany is",
    "locality_ground_truth": "Berlin",
    "subject": "the United Kingdom"
  },
  {
    "prompt": "Who is the current president of Argentina?",
    "target_new": "Javier Milei",
    "ground_truth": "Alberto Fernandez",
    "rephrase_prompt": "Who currently governs Argentina?",
    "locality_prompt": "The capital of China is",
    "locality_ground_truth": "Beijing",
    "subject": "Argentina"
  },
  {
    "prompt": "Who is the current monarch of the United Kingdom?",
    "target_new": "King Charles III",
    "ground_truth": "Queen Elizabeth II",
    "rephrase_prompt": "Who is the king of the UK now?",
    "locality_prompt": "The capital of Italy is",
    "locality_ground_truth": "Rome",
    "subject": "the United Kingdom"
  },
  {
    "prompt": "Who is the current CEO of YouTube?",
    "target_new": "Neal Mohan",
    "ground_truth": "Susan Wojcicki",
    "rephrase_prompt": "Who leads YouTube currently?",
    "locality_prompt": "The current CEO of Google is",
    "locality_ground_truth": "Sundar Pichai",
    "subject": "YouTube"
  },
  {
```

```
    "prompt": "Who is the current CEO of Starbucks?",
    "target_new": "Laxman Narasimhan",
    "ground_truth": "Howard Schultz",
    "rephrase_prompt": "Who is currently managing Starbucks?",
    "locality_prompt": "The current CEO of Apple is",
    "locality_ground_truth": "Tim Cook",
    "subject": "Starbucks"
  }
]
```

(2)编写推理脚本：运行文件为 baseline.py，此处不展开展示代码。运行方法为在根目录下运行如下代码

```
python baseline.py
```

运行结果如下图所示：



```
Sample 3
--------------------------------------------------------------------------------
Prompt:
Who is the CEO of OpenAI?

Old Fact:
Mira Murati

Target New Fact:
Sam Altman

Model Response:
The CEO of OpenAI is Sam Altman.
#endregion

Contains Old Fact: False
Contains New Fact: True
```

```
================================================================================
Sample 4
--------------------------------------------------------------------------------
Prompt:
What is the capital of Kazakhstan?

Old Fact:
Nur-Sultan

Target New Fact:
Astana

Model Response:
The capital of Kazakhstan is Astana.anko
 Crimea

Contains Old Fact: False
Contains New Fact: True

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 5
--------------------------------------------------------------------------------
Prompt:
Who is the owner of Twitter (X)?

Old Fact:
Jack Dorsey

Target New Fact:
Elon Musk

Model Response:
The owner of Twitter is Jack Dorsey.
ponde
You

Contains Old Fact: True
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 6
--------------------------------------------------------------------------------
Prompt:
Who is the current Prime Minister of the United Kingdom?

Old Fact:
Rishi Sunak

Target New Fact:
Keir Starmer

Model Response:
The current Prime Minister of the United Kingdom is Theresa May.

Contains Old Fact: False
Contains New Fact: False
```

```
================================================================================
Sample 7
--------------------------------------------------------------------------------
Prompt:
Who is the current president of Argentina?

Old Fact:
Alberto Fernandez

Target New Fact:
Javier Milei

Model Response:
The current president of Argentina is Cristina Kirchner.
 Crimea

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 8
--------------------------------------------------------------------------------
Prompt:
Who is the current monarch of the United Kingdom?

Old Fact:
Queen Elizabeth II

Target New Fact:
King Charles III

Model Response:
The current monarch of the United Kingdom is Queen Elizabeth II.

Contains Old Fact: True
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 9
--------------------------------------------------------------------------------
Prompt:
Who is the current CEO of YouTube?

Old Fact:
Susan Wojcicki

Target New Fact:
Neal Mohan

Model Response:
The current CEO of YouTube is Sheryl Sandberg.
离退休

Contains Old Fact: False
Contains New Fact: False
```

```
================================================================================
Sample 10
--------------------------------------------------------------------------------
Prompt:
Who is the current CEO of Starbucks?

Old Fact:
Howard Schultz

Target New Fact:
Laxman Narasimhan

Model Response:
The current CEO of Starbucks is Howard Schultz. Howard Schultz is

Contains Old Fact: True
Contains New Fact: False
```

```
================================================================================
BASELINE SUMMARY
================================================================================
Total Samples: 10
Old Fact Accuracy: 3/10 (30.00%)
New Fact Accuracy: 3/10 (30.00%)
```

实验结果显示仅有 3 条符合期望的答案，3 条回复符合旧知识，其余的为错误结果。可以证明模型在编辑前确实存在知识盲区或旧知识。

3.2 Task2: 单条事实编辑实践 (Single Fact Editing using ROME)

ROME (Rank-One Model Editing) 是一种经典的通过修改前馈神经网络（FFN）特定层权重来实现单条知识编辑的方法。

3.2.1 任务内容

(1)基于 EasyEdit 框架，配置 ROME 算法参数。

(2)将 Task 1 中的 10 条事实逐一进行编辑（每次重置模型权重）。

(3)验证：输入提示词“推特现任 CEO 是”，观察模型是否输出“Linda Yaccarino”。

3.2.2 实验结果与分析

(1) 修改 EasyEdit 框架使其与 Qwen2.5-0.5B 互相适配，需要在原始 EasyEdit/hparams/ROME 目录下添加配置文件 qwen2.5-0.5b.yaml（该仓库已配置好，无需重复配置），文件具体内容如下：

```yaml
alg_name: "ROME"
model_name: "Qwen/Qwen2.5-0.5B"
stats_dir: "./easyedit_data/stats"
device: 0
layers: [11]
fact_token: "subject_last"
v_num_grad_steps: 20
v_lr: 5e-1
```

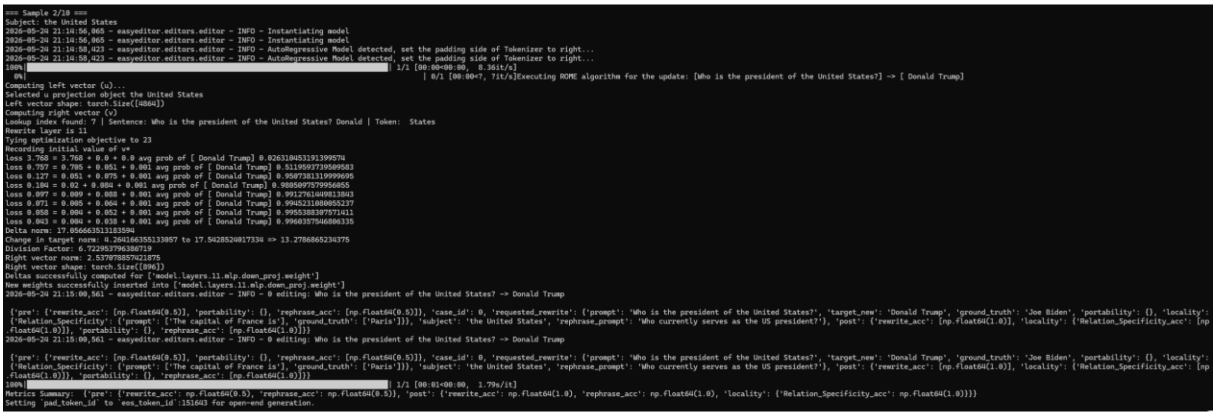
```yaml
v_loss_layer: 23
v_weight_decay: 1e-3
clamp_norm_factor: 4
kl_factor: 0.0625
mom2_adjustment: false
context_template_length_params: [[5, 10], [10, 10]]
rewrite_module_tmp: "model.layers.{}.mlp.down_proj"
layer_module_tmp: "model.layers.{}"
mlp_module_tmp: "model.layers.{}.mlp"
attn_module_tmp: "model.layers.{}.self_attn"
ln_f_module: "model.norm"
lm_head_module: "lm_head"
mom2_dataset: "wikipedia"
mom2_n_samples: 100000
mom2_dtype: "float32"
max_length: 128
model_parallel: false
fp16: false
```

(2)编写脚本进行知识编辑，脚本为根目录下的 edit_rome.py 文件，在根目录下执行以下代码即可运行

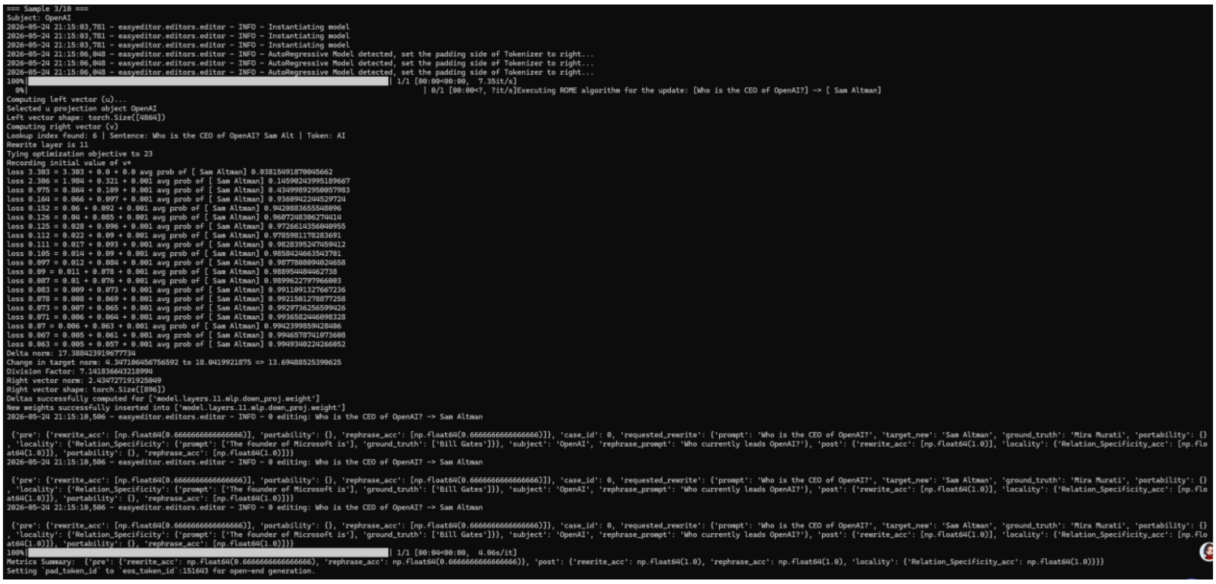
```
python edit_rome.py
```

(3)实验结果验证与分析

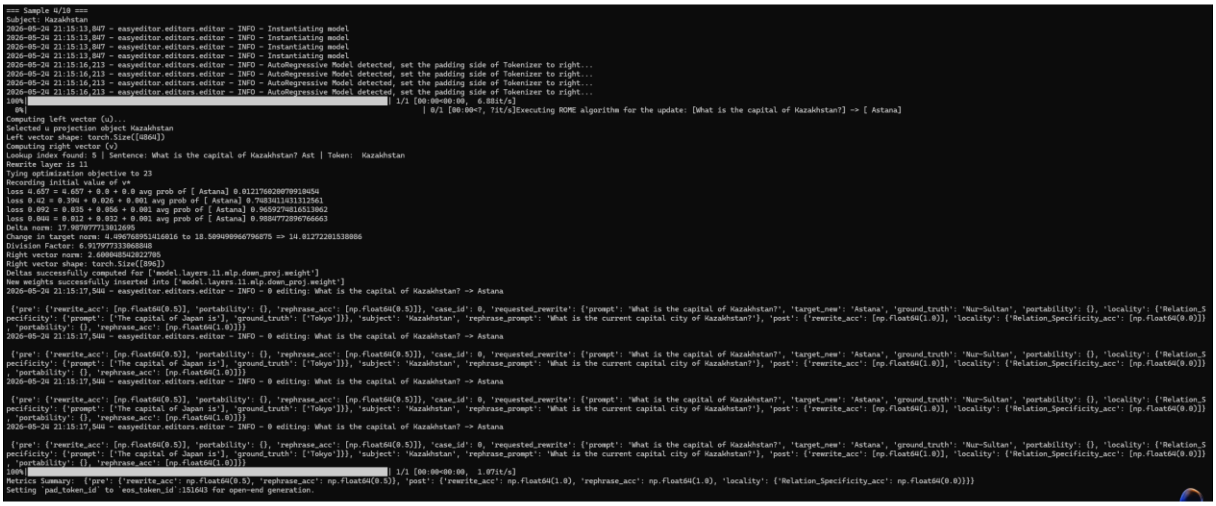
以下为运行截图



```
=== Sample 2/10 ===
Subject: the United States
2026-05-24 21:14:56,065 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:14:56,065 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:14:58,423 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:14:58,423 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
100%|██████████| 1/1 [00:00<00:00,  8.36it/s]
  0%|          | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [Who is the president of the United States?] -> [ Donald Trump]
Computing left vector (u)...
Selected u projection object the United States
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 7 | Sentence: Who is the president of the United States? Donald | Token:  States
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 3.768 = 3.768 + 0.0 + 0.0 avg prob of [ Donald Trump] 0.026310453191399574
loss 0.757 = 0.705 + 0.051 + 0.001 avg prob of [ Donald Trump] 0.5119593739509583
loss 0.127 = 0.051 + 0.075 + 0.001 avg prob of [ Donald Trump] 0.9507381319999695
loss 0.104 = 0.02 + 0.084 + 0.001 avg prob of [ Donald Trump] 0.9805097579956055
loss 0.097 = 0.009 + 0.088 + 0.001 avg prob of [ Donald Trump] 0.9912761449813843
loss 0.071 = 0.005 + 0.064 + 0.001 avg prob of [ Donald Trump] 0.9945231080055237
loss 0.058 = 0.004 + 0.052 + 0.001 avg prob of [ Donald Trump] 0.9955388307571411
loss 0.043 = 0.004 + 0.038 + 0.001 avg prob of [ Donald Trump] 0.9960357546806335
Delta norm: 17.056663513183594
Change in target norm: 4.264166355133057 to 17.5428524017334 => 13.2786865234375
Division Factor: 6.722953796386719
Right vector norm: 2.537078857421875
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.weight']
2026-05-24 21:15:00,561 - easyeditor.editors.editor - INFO - 0 editing: Who is the president of the United States? -> Donald Trump

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the president of the United States?', 'target_new': 'Donald Trump', 'ground_truth': 'Joe Biden', 'portability': {}, 'locality':
 {'Relation_Specificity': {'prompt': ['The capital of France is'], 'ground_truth': ['Paris']}}, 'subject': 'the United States', 'rephrase_prompt': 'Who currently serves as the US president?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np
.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:00,561 - easyeditor.editors.editor - INFO - 0 editing: Who is the president of the United States? -> Donald Trump

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the president of the United States?', 'target_new': 'Donald Trump', 'ground_truth': 'Joe Biden', 'portability': {}, 'locality':
 {'Relation_Specificity': {'prompt': ['The capital of France is'], 'ground_truth': ['Paris']}}, 'subject': 'the United States', 'rephrase_prompt': 'Who currently serves as the US president?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np
.float64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
100%|██████████| 1/1 [00:01<00:00,  1.79s/it]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.5), 'rephrase_acc': np.float64(0.5)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_acc': np.float64(1.0), 'locality': {'Relation_Specificity_acc': np.float64(1.0)}}}
Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
```



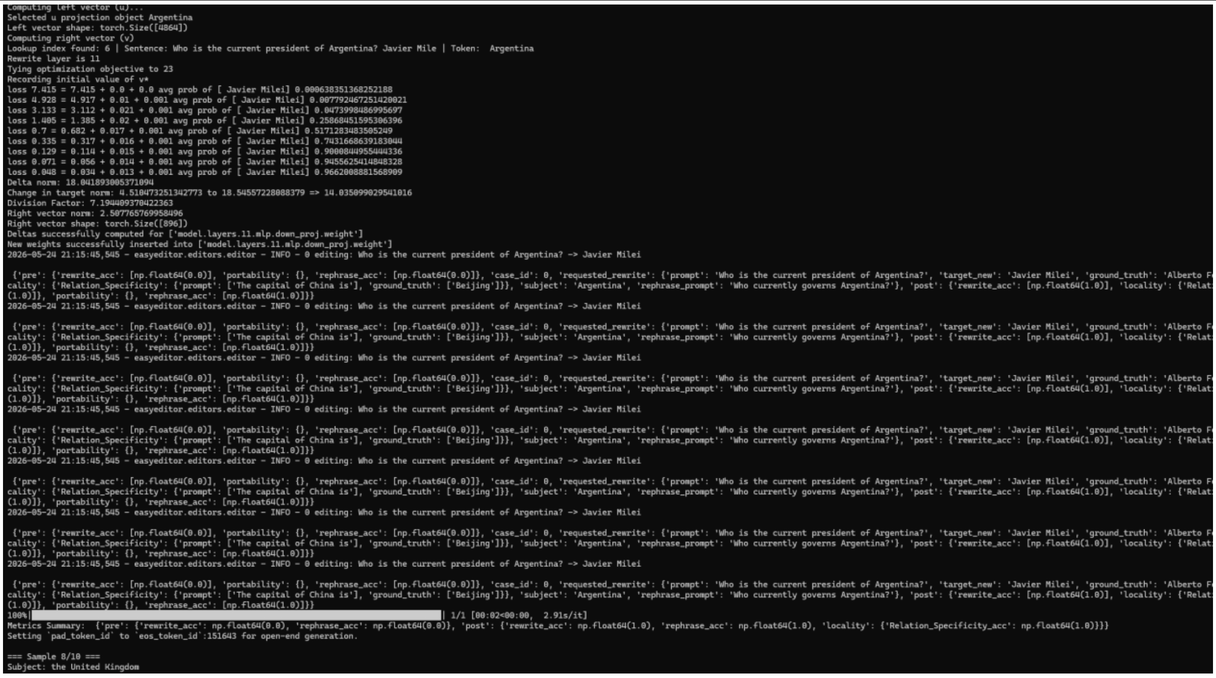
```
=== Sample 3/10 ===
Subject: OpenAI
2026-05-24 21:15:03,781 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:03,781 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:03,781 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:06,048 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:06,048 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:06,048 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
100%|██████████| 1/1 [00:00<00:00,  7.35it/s]
  0%|          | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [Who is the CEO of OpenAI?] -> [ Sam Altman]
Computing left vector (u)...
Selected u projection object OpenAI
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 6 | Sentence: Who is the CEO of OpenAI? Sam Alt | Token: AI
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 3.303 = 3.303 + 0.0 + 0.0 avg prob of [ Sam Altman] 0.03815491870045662
loss 2.306 = 1.984 + 0.321 + 0.001 avg prob of [ Sam Altman] 0.14590243995189667
loss 0.975 = 0.864 + 0.109 + 0.001 avg prob of [ Sam Altman] 0.43499892950057983
loss 0.164 = 0.066 + 0.097 + 0.001 avg prob of [ Sam Altman] 0.9360942244529724
loss 0.152 = 0.06 + 0.092 + 0.001 avg prob of [ Sam Altman] 0.9420883655548096
loss 0.126 = 0.04 + 0.085 + 0.001 avg prob of [ Sam Altman] 0.9607248306274414
loss 0.125 = 0.028 + 0.096 + 0.001 avg prob of [ Sam Altman] 0.9726614356040955
loss 0.112 = 0.022 + 0.09 + 0.001 avg prob of [ Sam Altman] 0.9785981178283691
loss 0.111 = 0.017 + 0.093 + 0.001 avg prob of [ Sam Altman] 0.9828395247459412
loss 0.105 = 0.014 + 0.09 + 0.001 avg prob of [ Sam Altman] 0.9858424663543701
loss 0.097 = 0.012 + 0.084 + 0.001 avg prob of [ Sam Altman] 0.9877808094024658
loss 0.09 = 0.011 + 0.078 + 0.001 avg prob of [ Sam Altman] 0.988954484462738
loss 0.087 = 0.01 + 0.076 + 0.001 avg prob of [ Sam Altman] 0.9899622797966003
loss 0.083 = 0.009 + 0.073 + 0.001 avg prob of [ Sam Altman] 0.9911091327667236
loss 0.078 = 0.008 + 0.069 + 0.001 avg prob of [ Sam Altman] 0.9921501278877258
loss 0.073 = 0.007 + 0.065 + 0.001 avg prob of [ Sam Altman] 0.9929736256599426
loss 0.071 = 0.006 + 0.064 + 0.001 avg prob of [ Sam Altman] 0.9936582446098328
loss 0.07 = 0.006 + 0.063 + 0.001 avg prob of [ Sam Altman] 0.9942399859428406
loss 0.067 = 0.005 + 0.061 + 0.001 avg prob of [ Sam Altman] 0.9946578741073608
loss 0.063 = 0.005 + 0.057 + 0.001 avg prob of [ Sam Altman] 0.9949340224266052
Delta norm: 17.388423919677734
Change in target norm: 4.347106456756592 to 18.0419921875 => 13.69488525390625
Division Factor: 7.141836643218994
Right vector norm: 2.434727191925049
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.weight']
2026-05-24 21:15:10,506 - easyeditor.editors.editor - INFO - 0 editing: Who is the CEO of OpenAI? -> Sam Altman

 {'pre': {'rewrite_acc': [np.float64(0.6666666666666666)], 'portability': {}, 'rephrase_acc': [np.float64(0.6666666666666666)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the CEO of OpenAI?', 'target_new': 'Sam Altman', 'ground_truth': 'Mira Murati', 'portability': {}
, 'locality': {'Relation_Specificity': {'prompt': ['The founder of Microsoft is'], 'ground_truth': ['Bill Gates']}}, 'subject': 'OpenAI', 'rephrase_prompt': 'Who currently leads OpenAI?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.flo
at64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:10,506 - easyeditor.editors.editor - INFO - 0 editing: Who is the CEO of OpenAI? -> Sam Altman

 {'pre': {'rewrite_acc': [np.float64(0.6666666666666666)], 'portability': {}, 'rephrase_acc': [np.float64(0.6666666666666666)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the CEO of OpenAI?', 'target_new': 'Sam Altman', 'ground_truth': 'Mira Murati', 'portability': {}
, 'locality': {'Relation_Specificity': {'prompt': ['The founder of Microsoft is'], 'ground_truth': ['Bill Gates']}}, 'subject': 'OpenAI', 'rephrase_prompt': 'Who currently leads OpenAI?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.flo
at64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:10,506 - easyeditor.editors.editor - INFO - 0 editing: Who is the CEO of OpenAI? -> Sam Altman

 {'pre': {'rewrite_acc': [np.float64(0.6666666666666666)], 'portability': {}, 'rephrase_acc': [np.float64(0.6666666666666666)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the CEO of OpenAI?', 'target_new': 'Sam Altman', 'ground_truth': 'Mira Murati', 'portability': {}
, 'locality': {'Relation_Specificity': {'prompt': ['The founder of Microsoft is'], 'ground_truth': ['Bill Gates']}}, 'subject': 'OpenAI', 'rephrase_prompt': 'Who currently leads OpenAI?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.flo
at64(1.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
100%|██████████| 1/1 [00:04<00:00,  4.06s/it]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.6666666666666666), 'rephrase_acc': np.float64(0.6666666666666666)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_acc': np.float64(1.0), 'locality': {'Relation_Specificity_acc': np.float64(1.0)}}}
Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
```



```
=== Sample 4/10 ===
Subject: Kazakhstan
2026-05-24 21:15:13,847 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:13,847 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:13,847 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:13,847 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:16,213 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:16,213 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:16,213 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:16,213 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
100%|██████████| 1/1 [00:00<00:00,  6.88it/s]
  0%|          | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [What is the capital of Kazakhstan?] -> [ Astana]
Computing left vector (u)...
Selected u projection object Kazakhstan
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 5 | Sentence: What is the capital of Kazakhstan? Ast | Token:  Kazakhstan
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 4.657 = 4.657 + 0.0 + 0.0 avg prob of [ Astana] 0.012176028070910454
loss 0.42 = 0.394 + 0.026 + 0.001 avg prob of [ Astana] 0.7483411431312561
loss 0.092 = 0.035 + 0.056 + 0.001 avg prob of [ Astana] 0.9659274816513062
loss 0.044 = 0.012 + 0.032 + 0.001 avg prob of [ Astana] 0.9884772896766663
Delta norm: 17.987077713012695
Change in target norm: 4.496768951416016 to 18.509490966796875 => 14.01272201538086
Division Factor: 6.917977333068848
Right vector norm: 2.600048542022705
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.weight']
2026-05-24 21:15:17,544 - easyeditor.editors.editor - INFO - 0 editing: What is the capital of Kazakhstan? -> Astana

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'What is the capital of Kazakhstan?', 'target_new': 'Astana', 'ground_truth': 'Nur-Sultan', 'portability': {}, 'locality': {'Relation_S
pecificity': {'prompt': ['The capital of Japan is'], 'ground_truth': ['Tokyo']}}, 'subject': 'Kazakhstan', 'rephrase_prompt': 'What is the current capital city of Kazakhstan?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(0.0)]}
, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:17,544 - easyeditor.editors.editor - INFO - 0 editing: What is the capital of Kazakhstan? -> Astana

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'What is the capital of Kazakhstan?', 'target_new': 'Astana', 'ground_truth': 'Nur-Sultan', 'portability': {}, 'locality': {'Relation_S
pecificity': {'prompt': ['The capital of Japan is'], 'ground_truth': ['Tokyo']}}, 'subject': 'Kazakhstan', 'rephrase_prompt': 'What is the current capital city of Kazakhstan?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(0.0)]}
, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:17,544 - easyeditor.editors.editor - INFO - 0 editing: What is the capital of Kazakhstan? -> Astana

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'What is the capital of Kazakhstan?', 'target_new': 'Astana', 'ground_truth': 'Nur-Sultan', 'portability': {}, 'locality': {'Relation_S
pecificity': {'prompt': ['The capital of Japan is'], 'ground_truth': ['Tokyo']}}, 'subject': 'Kazakhstan', 'rephrase_prompt': 'What is the current capital city of Kazakhstan?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(0.0)]}
, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:17,544 - easyeditor.editors.editor - INFO - 0 editing: What is the capital of Kazakhstan? -> Astana

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'What is the capital of Kazakhstan?', 'target_new': 'Astana', 'ground_truth': 'Nur-Sultan', 'portability': {}, 'locality': {'Relation_S
pecificity': {'prompt': ['The capital of Japan is'], 'ground_truth': ['Tokyo']}}, 'subject': 'Kazakhstan', 'rephrase_prompt': 'What is the current capital city of Kazakhstan?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(0.0)]}
, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
100%|██████████| 1/1 [00:00<00:00,  1.07it/s]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.5), 'rephrase_acc': np.float64(0.5)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_acc': np.float64(1.0), 'locality': {'Relation_Specificity_acc': np.float64(0.0)}}}
Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
```

```
=== Sample 5/10 ===
Subject: Twitter (X)
2026-05-24 21:15:20,722 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:20,722 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:20,722 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:20,722 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:20,722 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:23,066 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:23,066 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:23,066 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:23,066 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:23,066 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
100%|██████████████████████████████████████████████████| 1/1 [00:00<00:00,  7.35it/s]
  0%|                                                  | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [Who is the owner of Twitter (X)?] -> [ Elon Musk]
Computing left vector (u)...
Selected u projection object Twitter (X)
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 8 | Sentence: Who is the owner of Twitter (X)? Elon | Token: )?
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 4.323 = 4.323 + 0.0 + 0.0 avg prob of [ Elon Musk] 0.014984434470534325
loss 0.983 = 0.859 + 0.123 + 0.001 avg prob of [ Elon Musk] 0.43220967054367065
loss 0.186 = 0.064 + 0.122 + 0.001 avg prob of [ Elon Musk] 0.9398815496444702
loss 0.114 = 0.037 + 0.077 + 0.001 avg prob of [ Elon Musk] 0.9642730355262756
loss 0.082 = 0.028 + 0.053 + 0.001 avg prob of [ Elon Musk] 0.9721201062202454
loss 0.073 = 0.025 + 0.047 + 0.001 avg prob of [ Elon Musk] 0.9754657149931488
loss 0.062 = 0.018 + 0.043 + 0.001 avg prob of [ Elon Musk] 0.9818903803825378
loss 0.05 = 0.012 + 0.037 + 0.001 avg prob of [ Elon Musk] 0.9885054230690002
Delta norm: 20.6861572265625
Change in target norm: 5.171539306640625 to 21.1732234954834 => 16.001684188842773
Division Factor: 7.962900161743164
Right vector norm: 2.5978169441223145
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.weight']
2026-05-24 21:15:25,298 - easyeditor.editors.editor - INFO - 0 editing: Who is the owner of Twitter (X)? -> Elon Musk

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the owner of Twitter (X)?', 'target_new': 'Elon Musk', 'ground_truth': 'Jack Dorsey', 'portability': {}, 'locality': {'Relation
_Specificity': {'prompt': ['The CEO of Tesla is'], 'ground_truth': ['Elon Musk']}}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who owns X platform currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(1.0)]}, 'portability
': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:25,298 - easyeditor.editors.editor - INFO - 0 editing: Who is the owner of Twitter (X)? -> Elon Musk

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the owner of Twitter (X)?', 'target_new': 'Elon Musk', 'ground_truth': 'Jack Dorsey', 'portability': {}, 'locality': {'Relation
_Specificity': {'prompt': ['The CEO of Tesla is'], 'ground_truth': ['Elon Musk']}}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who owns X platform currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(1.0)]}, 'portability
': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:25,298 - easyeditor.editors.editor - INFO - 0 editing: Who is the owner of Twitter (X)? -> Elon Musk

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the owner of Twitter (X)?', 'target_new': 'Elon Musk', 'ground_truth': 'Jack Dorsey', 'portability': {}, 'locality': {'Relation
_Specificity': {'prompt': ['The CEO of Tesla is'], 'ground_truth': ['Elon Musk']}}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who owns X platform currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(1.0)]}, 'portability
': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:25,298 - easyeditor.editors.editor - INFO - 0 editing: Who is the owner of Twitter (X)? -> Elon Musk

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the owner of Twitter (X)?', 'target_new': 'Elon Musk', 'ground_truth': 'Jack Dorsey', 'portability': {}, 'locality': {'Relation
_Specificity': {'prompt': ['The CEO of Tesla is'], 'ground_truth': ['Elon Musk']}}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who owns X platform currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(1.0)]}, 'portability
': {}, 'rephrase_acc': [np.float64(1.0)]}}
2026-05-24 21:15:25,298 - easyeditor.editors.editor - INFO - 0 editing: Who is the owner of Twitter (X)? -> Elon Musk

 {'pre': {'rewrite_acc': [np.float64(0.5)], 'portability': {}, 'rephrase_acc': [np.float64(0.5)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the owner of Twitter (X)?', 'target_new': 'Elon Musk', 'ground_truth': 'Jack Dorsey', 'portability': {}, 'locality': {'Relation
_Specificity': {'prompt': ['The CEO of Tesla is'], 'ground_truth': ['Elon Musk']}}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who owns X platform currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.float64(1.0)]}, 'portability
': {}, 'rephrase_acc': [np.float64(1.0)]}}
100%|██████████████████████████████████████████████████| 1/1 [00:01<00:00,  1.83s/it]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.5), 'rephrase_acc': np.float64(0.5)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_acc': np.float64(1.0), 'locality': {'Relation_Specificity_acc': np.float64(1.0)}}}
Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
```

```
=== Sample 6/10 ===
Subject: the United Kingdom
2026-05-24 21:15:28,633 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:28,633 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:28,633 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:28,633 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:28,633 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:28,633 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 21:15:30,928 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:30,928 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:30,928 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:30,928 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:30,928 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
2026-05-24 21:15:30,928 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
100%|██████████████████████████████████████████████████| 1/1 [00:00<00:00,  7.10it/s]
  0%|                                                  | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [Who is the current Prime Minister of the United Kingdom?] -> [ Keir Starmer]
Computing left vector (u)...
Selected u projection object the United Kingdom
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 9 | Sentence: Who is the current Prime Minister of the United Kingdom? Keir Star | Token:  Kingdom
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 2.613 = 2.613 + 0.0 + 0.0 avg prob of [ Keir Starmer] 0.07445385307073593
loss 1.337 = 1.318 + 0.018 + 0.001 avg prob of [ Keir Starmer] 0.2711792588233948
loss 0.718 = 0.672 + 0.045 + 0.001 avg prob of [ Keir Starmer] 0.5141991972923279
loss 0.141 = 0.096 + 0.045 + 0.001 avg prob of [ Keir Starmer] 0.9104180932044983
loss 0.114 = 0.086 + 0.027 + 0.001 avg prob of [ Keir Starmer] 0.9182340502738953
loss 0.086 = 0.063 + 0.022 + 0.001 avg prob of [ Keir Starmer] 0.9389304518699646
loss 0.068 = 0.048 + 0.019 + 0.001 avg prob of [ Keir Starmer] 0.9532248973846436
loss 0.052 = 0.037 + 0.014 + 0.001 avg prob of [ Keir Starmer] 0.9637882113456726
loss 0.042 = 0.029 + 0.012 + 0.001 avg prob of [ Keir Starmer] 0.9717636108398438
Delta norm: 17.320417404174805
Change in target norm: 4.330104827880859 to 17.83913230895996 => 13.509027481079102
Division Factor: 7.094655513763428
Right vector norm: 2.441333293914795
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.weight']
2026-05-24 21:15:36,482 - easyeditor.editors.editor - INFO - 0 editing: Who is the current Prime Minister of the United Kingdom? -> Keir Starmer

 {'pre': {'rewrite_acc': [np.float64(0.75)], 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the current Prime Minister of the United Kingdom?', 'target_new': 'Keir Starmer', 'ground_truth': 'Rishi Sunak', 'portabi
': {}, 'locality': {'Relation_Specificity': {'prompt': ['The capital of Germany is'], 'ground_truth': ['Berlin']}}, 'subject': 'the United Kingdom', 'rephrase_prompt': 'Who is serving as the UK Prime Minister currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality'
Relation_Specificity_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}}
2026-05-24 21:15:36,482 - easyeditor.editors.editor - INFO - 0 editing: Who is the current Prime Minister of the United Kingdom? -> Keir Starmer

 {'pre': {'rewrite_acc': [np.float64(0.75)], 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the current Prime Minister of the United Kingdom?', 'target_new': 'Keir Starmer', 'ground_truth': 'Rishi Sunak', 'portabi
': {}, 'locality': {'Relation_Specificity': {'prompt': ['The capital of Germany is'], 'ground_truth': ['Berlin']}}, 'subject': 'the United Kingdom', 'rephrase_prompt': 'Who is serving as the UK Prime Minister currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality'
Relation_Specificity_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}}
2026-05-24 21:15:36,482 - easyeditor.editors.editor - INFO - 0 editing: Who is the current Prime Minister of the United Kingdom? -> Keir Starmer

 {'pre': {'rewrite_acc': [np.float64(0.75)], 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the current Prime Minister of the United Kingdom?', 'target_new': 'Keir Starmer', 'ground_truth': 'Rishi Sunak', 'portabi
': {}, 'locality': {'Relation_Specificity': {'prompt': ['The capital of Germany is'], 'ground_truth': ['Berlin']}}, 'subject': 'the United Kingdom', 'rephrase_prompt': 'Who is serving as the UK Prime Minister currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality'
Relation_Specificity_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}}
2026-05-24 21:15:36,482 - easyeditor.editors.editor - INFO - 0 editing: Who is the current Prime Minister of the United Kingdom? -> Keir Starmer

 {'pre': {'rewrite_acc': [np.float64(0.75)], 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the current Prime Minister of the United Kingdom?', 'target_new': 'Keir Starmer', 'ground_truth': 'Rishi Sunak', 'portabi
': {}, 'locality': {'Relation_Specificity': {'prompt': ['The capital of Germany is'], 'ground_truth': ['Berlin']}}, 'subject': 'the United Kingdom', 'rephrase_prompt': 'Who is serving as the UK Prime Minister currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality'
Relation_Specificity_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}}
2026-05-24 21:15:36,482 - easyeditor.editors.editor - INFO - 0 editing: Who is the current Prime Minister of the United Kingdom? -> Keir Starmer

 {'pre': {'rewrite_acc': [np.float64(0.75)], 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the current Prime Minister of the United Kingdom?', 'target_new': 'Keir Starmer', 'ground_truth': 'Rishi Sunak', 'portabi
': {}, 'locality': {'Relation_Specificity': {'prompt': ['The capital of Germany is'], 'ground_truth': ['Berlin']}}, 'subject': 'the United Kingdom', 'rephrase_prompt': 'Who is serving as the UK Prime Minister currently?'}, 'post': {'rewrite_acc': [np.float64(1.0)], 'locality'
Relation_Specificity_acc': [np.float64(0.0)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}}
2026-05-24 21:15:36,482 - easyeditor.editors.editor - INFO - 0 editing: Who is the current Prime Minister of the United Kingdom? -> Keir Starmer

 {'pre': {'rewrite_acc': [np.float64(0.75)], 'portability': {}, 'rephrase_acc': [np.float64(0.75)]}, 'case_id': 0, 'requested_rewrite': {'prompt': 'Who is the current Prime Minister of the United Kingdom?', 'target_new': 'Keir Starmer', 'ground_truth': 'Rishi Sunak', 'portabi
```



十个样本均运行完毕，选取其中一个结果进行具体分析。以样本一知识编辑结果为例，样本一输出如下：

```
=== Sample 1/10 ===
Subject: Twitter (X)
2026-05-24 21:14:46,211 - easyeditor.editors.editor - INFO - Instantiatin
g model
`torch_dtype` is deprecated! Use `dtype` instead!
2026-05-24 21:14:48,484 - easyeditor.editors.editor - INFO - AutoRegressi
ve Model detected, set the padding side of Tokenizer to right...
100%|████████████████████████████████████████
███████████████████████████████████████████
█| 1/1 [00:00<00:00,  3.83it/s]
  0%|
          | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update:
```

```
[Who is the current CEO of Twitter (X)?] -> [ Linda Yaccarino]
Cached context templates ['{}', 'The following isosaurus. {}', 'The follo
wing are you can. {}', 'Therefore, I come from. {}', 'Therefore, it isom.
 {}', 'Because it has the the. {}', 'Because there is it is. {}', "I'm11\
n. {}", 'I am a = . {}', 'You are a will write. {}', 'You are an agent wil
l. {}', 'The Purple River of ( ) inductive,2. {}', 'The following isosaur
usine of orogen of. {}', "Therefore the function. I come National Justiça
's why. {}", 'Therefore, the function, it is the maximum of. {}', 'Becaus
e it is it is a man the number of. {}', 'Because of the of the the number
of its the. {}', 'I have  am a, a.  am. {}', "I'm  am Aleint 28. {}", "You
are in your优质回答了I've. {}", 'You are in a are a are a are a. {}']
Computing left vector (u)...
Selected u projection object Twitter (X)
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 9 | Sentence: Who is the current CEO of Twitter (X)? L
inda Yaccar | Token: )?
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 5.73 = 5.73 + 0.0 + 0.0 avg prob of [ Linda Yaccarino] 0.003310911823
064089
loss 4.505 = 4.413 + 0.092 + 0.001 avg prob of [ Linda Yaccarino] 0.012264
161370694637
loss 4.698 = 4.583 + 0.114 + 0.001 avg prob of [ Linda Yaccarino] 0.010553
505271673203
loss 3.563 = 3.488 + 0.075 + 0.001 avg prob of [ Linda Yaccarino] 0.030782
438814640045
loss 3.128 = 3.055 + 0.072 + 0.001 avg prob of [ Linda Yaccarino] 0.047367
29711294174
loss 2.373 = 2.313 + 0.059 + 0.001 avg prob of [ Linda Yaccarino] 0.099623
10642004013
loss 1.467 = 1.396 + 0.07 + 0.001 avg prob of [ Linda Yaccarino] 0.2507833
242416382
loss 0.703 = 0.61 + 0.092 + 0.001 avg prob of [ Linda Yaccarino] 0.5508739
948272705
loss 0.33 = 0.246 + 0.083 + 0.001 avg prob of [ Linda Yaccarino] 0.7861565
351486206
loss 0.131 = 0.069 + 0.061 + 0.001 avg prob of [ Linda Yaccarino] 0.933952
3315429688
loss 0.109 = 0.057 + 0.052 + 0.001 avg prob of [ Linda Yaccarino] 0.946312
665939331
loss 0.098 = 0.049 + 0.048 + 0.001 avg prob of [ Linda Yaccarino] 0.953539
8483276367
loss 0.087 = 0.042 + 0.044 + 0.001 avg prob of [ Linda Yaccarino] 0.959397
```

```
6736068726
loss 0.067 = 0.029 + 0.037 + 0.001 avg prob of [ Linda Yaccarino] 0.971714
3774032593
loss 0.051 = 0.018 + 0.032 + 0.001 avg prob of [ Linda Yaccarino] 0.981784
999370575
loss 0.042 = 0.013 + 0.028 + 0.001 avg prob of [ Linda Yaccarino] 0.987149
9538421631
Delta norm: 20.57695198059082
Change in target norm: 5.144237995147705 to 21.267230987548828 => 16.1229
9346923828
Division Factor: 7.869274139404297
Right vector norm: 2.614847421646118
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight
']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.w
eight']
2026-05-24 21:14:52,969 - easyeditor.editors.editor - INFO - 0 editing: W
ho is the current CEO of Twitter (X)? -> Linda Yaccarino

 {'pre': {'rewrite_acc': [np.float64(0.4)], 'portability': {}, 'rephrase
_acc': [np.float64(0.4)]}, 'case_id': 0, 'requested_rewrite': {'prompt':
 'Who is the current CEO of Twitter (X)?', 'target_new': 'Linda Yaccarino
', 'ground_truth': 'Elon Musk', 'portability': {}, 'locality': {'Relation
_Specificity': {'prompt': ['The current CEO of Apple is'], 'ground_truth
': ['Tim Cook']}}, 'subject': 'Twitter (X)', 'rephrase_prompt': 'Who is t
he chief executive officer of X formerly Twitter?'}, 'post': {'rewrite_ac
c': [np.float64(1.0)], 'locality': {'Relation_Specificity_acc': [np.floa
t64(0.5)]}, 'portability': {}, 'rephrase_acc': [np.float64(1.0)]}}
100%|████████████████████████████████████████
███████████████████████████████████████████
█| 1/1 [00:03<00:00,  3.99s/it]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.4), 'rephrase_acc
': np.float64(0.4)}, 'post': {'rewrite_acc': np.float64(1.0), 'rephrase_
acc': np.float64(1.0), 'locality': {'Relation_Specificity_acc': np.float
64(0.5)}}}
```

本次编辑初始loss为5.73，目标答案预测概率0.0033，迭代结束后loss下降至0.042，预测概率提升至0.9871，损失明显回落且模型置信度大幅上涨，过程中顺利算出参数增量并将权重更新至模型第 11 层 MLP 模块，参数实现实际修改，评测数据显示编辑前 rewrite_acc 与 rephrase_acc 均为 0.4，编辑后两项数值均提升至 1.0，改写效果符合预期，无关知识留存数值为 0.5，存在轻微的知识扰动情况，整体本次知识编辑任务完成度较高，改写效果表现良好。

3.3 Task3：批量知识编辑实践（Batch Editing using MEMIT）

MEMIT 算法在 ROME 的基础上进行了扩展，支持一次性向模型中注入成百上千条知识，同时保持较低的性能损耗。

3.3.1 任务内容

(1)选取开源知识编辑数据集 ZsRE 或 CounterFact 等其他数据集中的 500 条数据作为批量编辑集。

(2)使用 MEMIT 算法对模型进行一次性批量知识注入。

(3)记录批量编辑过程的显存占用和耗时情况。

3.3.2 实验结果与分析

(1) 随机选取 ZsRE 数据集的 500 条数据，实现文件为 sample_zsre.py。选取之后的数据集存放在根目录下的 data 目录下 ZsRE.json。

(2) MEMIT 算法使用与 ROMR 算法一样需要添加参数文件，路径为 \EasyEdit\hparams\MEMIT\qwen2.5-0.5b.yaml。文件配置如下：

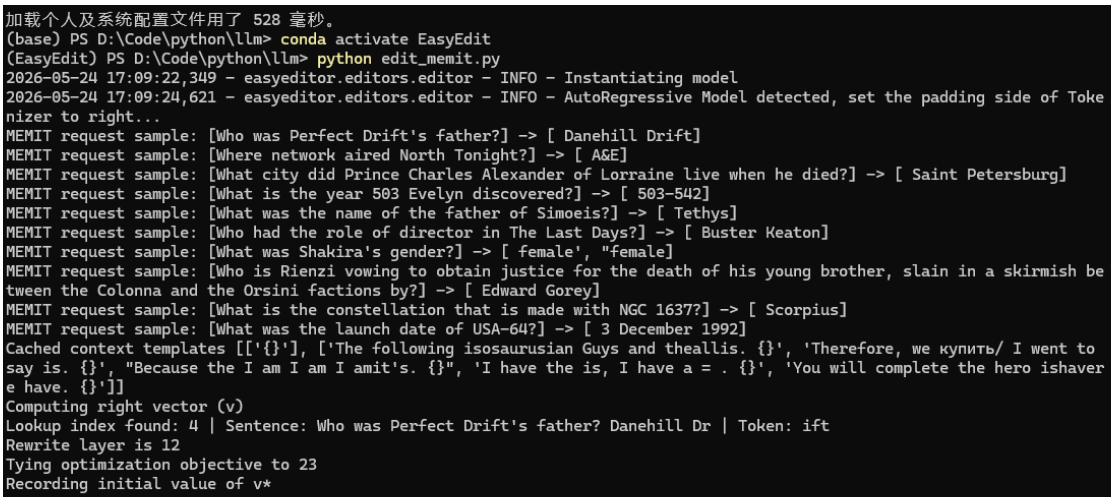
```
alg_name: "MEMIT"
model_name: "Qwen/Qwen2.5-0.5B"
stats_dir: "./easyedit_data/stats"
device: 0
layers: [8, 9, 10, 11, 12]
clamp_norm_factor: 4
layer_selection: "all"
fact_token: "subject_last"
v_num_grad_steps: 20
v_lr: 5e-1
v_loss_layer: 23
v_weight_decay: 1e-3
kl_factor: 0.0625
mom2_adjustment: true
mom2_update_weight: 15000
rewrite_module_tmp: "model.layers.{}.mlp.down_proj"
layer_module_tmp: "model.layers.{}"
mlp_module_tmp: "model.layers.{}.mlp"
attn_module_tmp: "model.layers.{}.self_attn"
ln_f_module: "model.norm"
lm_head_module: "model.embed_tokens"
mom2_dataset: "./data/knowedit/ZsRE/zsre_stats_corpus.jsonl"
mom2_n_samples: 5000
```

```
mom2_dtype: "float32"
max_length: 128
batch_size: 1
model_parallel: false
```

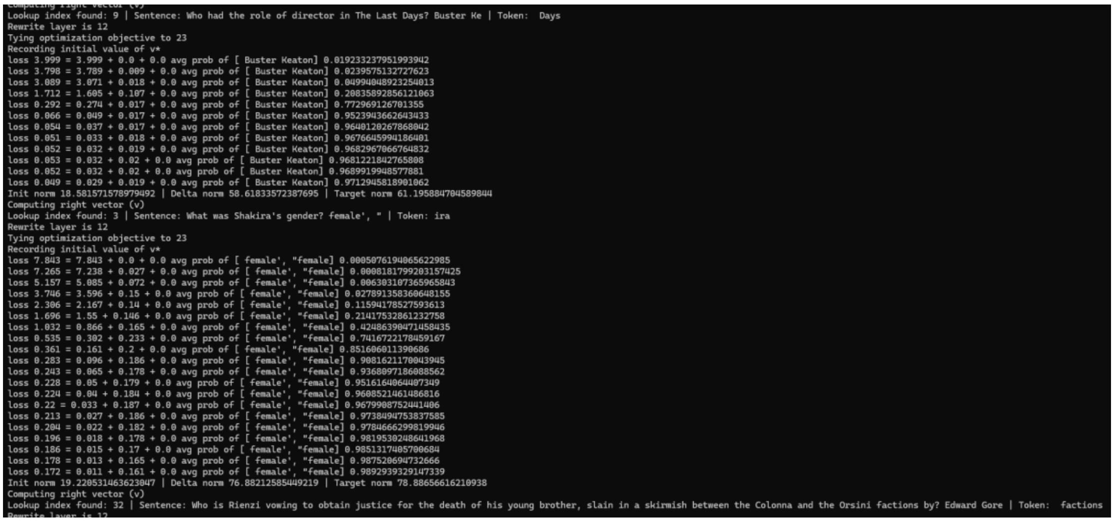
之后在根目录下运行如下代码：

```
python edit_memit.py
```

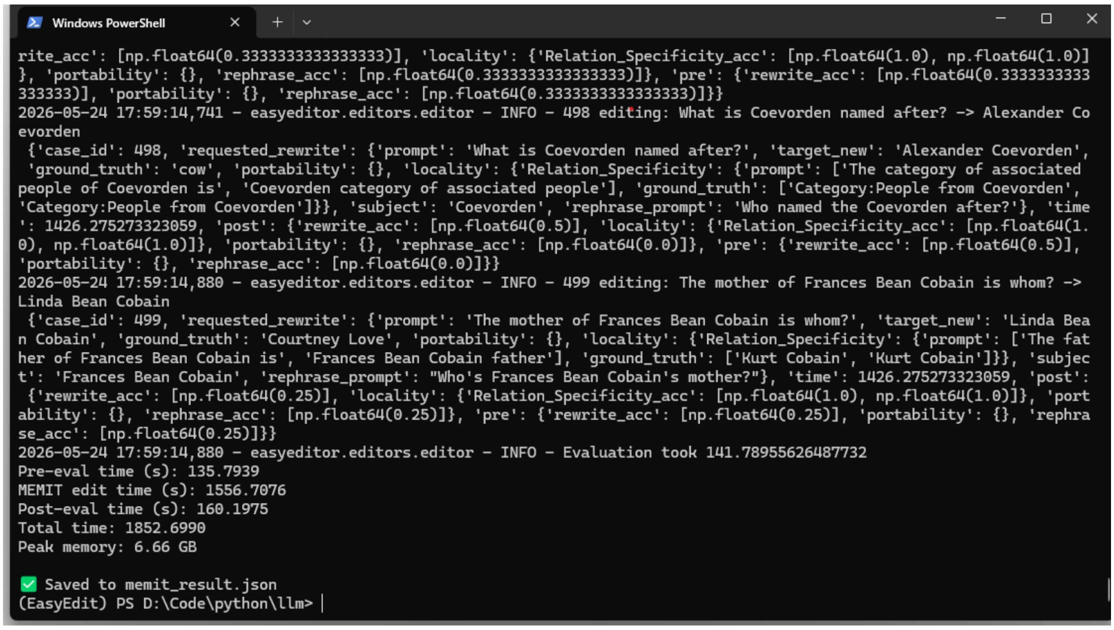
(3)以下为运行结果截图



```
加载个人及系统配置文件用了 528 毫秒。
(base) PS D:\Code\python\llm> conda activate EasyEdit
(EasyEdit) PS D:\Code\python\llm> python edit_memit.py
2026-05-24 17:09:22,349 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 17:09:24,621 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Toke
nizer to right...
MEMIT request sample: [Who was Perfect Drift's father?] -> [ Danehill Drift]
MEMIT request sample: [Where network aired North Tonight?] -> [ A&E]
MEMIT request sample: [What city did Prince Charles Alexander of Lorraine live when he died?] -> [ Saint Petersburg]
MEMIT request sample: [What is the year 503 Evelyn discovered?] -> [ 503-542]
MEMIT request sample: [What was the name of the father of Simoeis?] -> [ Tethys]
MEMIT request sample: [Who had the role of director in The Last Days?] -> [ Buster Keaton]
MEMIT request sample: [What was Shakira's gender?] -> [ female', "female]
MEMIT request sample: [Who is Rienzi vowing to obtain justice for the death of his young brother, slain in a skirmish be
tween the Colonna and the Orsini factions by?] -> [ Edward Gorey]
MEMIT request sample: [What is the constellation that is made with NGC 1637?] -> [ Scorpius]
MEMIT request sample: [What was the launch date of USA-64?] -> [ 3 December 1992]
Cached context templates [['{}'], ['The following isosaurusian Guys and theallis. {}', 'Therefore, we купить/ I went to
say is. {}', "Because the I am I am I amit's. {}", 'I have the is, I have a = . {}', 'You will complete the hero ishaver
e have. {}']]
Computing right vector (v)
Lookup index found: 4 | Sentence: Who was Perfect Drift's father? Danehill Dr | Token: ift
Rewrite layer is 12
Tying optimization objective to 23
Recording initial value of v*
```



```
Computing right vector (v)
Lookup index found: 9 | Sentence: Who had the role of director in The Last Days? Buster Ke | Token:  Days
Rewrite layer is 12
Tying optimization objective to 23
Recording initial value of v*
loss 3.999 = 3.999 + 0.0 + 0.0 avg prob of [ Buster Keaton] 0.019233237951993942
loss 3.798 = 3.789 + 0.009 + 0.0 avg prob of [ Buster Keaton] 0.02395751327276 23
loss 3.089 = 3.071 + 0.018 + 0.0 avg prob of [ Buster Keaton] 0.04994048923254013
loss 1.712 = 1.605 + 0.107 + 0.0 avg prob of [ Buster Keaton] 0.20835892856121063
loss 0.292 = 0.274 + 0.017 + 0.0 avg prob of [ Buster Keaton] 0.772969126701355
loss 0.066 = 0.049 + 0.017 + 0.0 avg prob of [ Buster Keaton] 0.9523943662643433
loss 0.054 = 0.037 + 0.017 + 0.0 avg prob of [ Buster Keaton] 0.9640120267868042
loss 0.051 = 0.033 + 0.018 + 0.0 avg prob of [ Buster Keaton] 0.9676645994186401
loss 0.052 = 0.032 + 0.019 + 0.0 avg prob of [ Buster Keaton] 0.9682967066764832
loss 0.053 = 0.032 + 0.02 + 0.0 avg prob of [ Buster Keaton] 0.9681221842765808
loss 0.052 = 0.032 + 0.02 + 0.0 avg prob of [ Buster Keaton] 0.9689919948577881
loss 0.049 = 0.029 + 0.019 + 0.0 avg prob of [ Buster Keaton] 0.9712945818901062
Init norm 18.581571578979492 | Delta norm 58.61833572387695 | Target norm 61.195884704589844
Computing right vector (v)
Lookup index found: 3 | Sentence: What was Shakira's gender? female', " | Token: ira
Rewrite layer is 12
Tying optimization objective to 23
Recording initial value of v*
loss 7.843 = 7.843 + 0.0 + 0.0 avg prob of [ female', "female] 0.0005076194065622985
loss 7.265 = 7.238 + 0.027 + 0.0 avg prob of [ female', "female] 0.0008181799203157425
loss 5.157 = 5.085 + 0.072 + 0.0 avg prob of [ female', "female] 0.006303107365965843
loss 3.746 = 3.596 + 0.15 + 0.0 avg prob of [ female', "female] 0.027891358360648155
loss 2.306 = 2.167 + 0.14 + 0.0 avg prob of [ female', "female] 0.11594178527593613
loss 1.696 = 1.55 + 0.146 + 0.0 avg prob of [ female', "female] 0.21417532861232758
loss 1.032 = 0.866 + 0.165 + 0.0 avg prob of [ female', "female] 0.42486390471458435
loss 0.535 = 0.302 + 0.233 + 0.0 avg prob of [ female', "female] 0.7416722178459167
loss 0.361 = 0.161 + 0.2 + 0.0 avg prob of [ female', "female] 0.851606011390686
loss 0.283 = 0.096 + 0.186 + 0.0 avg prob of [ female', "female] 0.9081621170043945
loss 0.243 = 0.065 + 0.178 + 0.0 avg prob of [ female', "female] 0.9368097186088562
loss 0.228 = 0.05 + 0.179 + 0.0 avg prob of [ female', "female] 0.9516164064407349
loss 0.224 = 0.04 + 0.184 + 0.0 avg prob of [ female', "female] 0.9608521461486816
loss 0.22 = 0.033 + 0.187 + 0.0 avg prob of [ female', "female] 0.9679908752441406
loss 0.213 = 0.027 + 0.186 + 0.0 avg prob of [ female', "female] 0.9738494753837585
loss 0.204 = 0.022 + 0.182 + 0.0 avg prob of [ female', "female] 0.9784666299819946
loss 0.196 = 0.018 + 0.178 + 0.0 avg prob of [ female', "female] 0.9819530248641968
loss 0.186 = 0.015 + 0.17 + 0.0 avg prob of [ female', "female] 0.9851317405700684
loss 0.178 = 0.013 + 0.165 + 0.0 avg prob of [ female', "female] 0.987520694732666
loss 0.172 = 0.011 + 0.161 + 0.0 avg prob of [ female', "female] 0.9892939329147339
Init norm 19.220531463623047 | Delta norm 76.88212585449219 | Target norm 78.88656616210938
Computing right vector (v)
Lookup index found: 32 | Sentence: Who is Rienzi vowing to obtain justice for the death of his young brother, slain in a skirmish between the Colonna and the Orsini factions by? Edward Gore | Token:  factions
Rewrite layer is 12
```



显存和耗时具体如下表所示：

表 1 显存和耗时记录

总耗时	MEMIT 编辑时间	显存最大占用空间
30.87min	1556.7076s	6.66GB

3.4 Task4：综合评估（Comprehensive Evaluation）

3.4.1 任务内容

计算以下三个核心指标：

(1)编辑成功率（Efficacy, ES）： 模型对直接编辑的 Prompt 是否输出了目标答案。

(2)泛化性（Generalization, PS）：模型对编辑事实的同义改写 Prompt 是否能输出目标答案。（例如：“谁目前在管理推特？”）

(3)局部性（Locality, NS）： 模型对无关事实的回答是否受到了破坏。（例如：编辑了推特的 CEO，模型对“苹果的 CEO 是谁”的回答是否依然正确）。

3.4.2 实验结果与分析

MEMIT 实验结果保存在根目录下 memit_result.json。对得到的结果进行计算评估，在根目录下执行以下命令即可：

```
python evaluate.py
```

运行结果如下图所示：

```
(EasyEdit) PS D:\Code\python\llm> python evaluate.py
================================================================================
Method: MEMIT
Model: Qwen/Qwen2.5-0.5B
Pre ES: 0.0540
Post ES: 0.2340
Pre PS: 0.0420
Post PS: 0.1800
Pre NS: 0.0780
Post NS: 0.0310
```

可列如下表格：

表 2 综合评估结果

评价指标	MEMIT 编辑前	MEMIT 编辑后
ES	0.054	0.234
PS	0.042	0.18
NS	0.078	0.031

从编辑成功率、泛化性和局部性三个角度来看，本次 MEMIT 知识编辑在模型性能上呈现出明显变化：编辑成功率（ES）从编辑前的 0.054 提升至 0.234，泛化性（PS）从 0.042 提升至 0.18，说明模型对直接编辑和同义改写的知识问答能力均得到增强，编辑效果有效传递到了不同表述的问题上；但局部性（NS）从 0.078 下降至 0.031，反映出编辑过程对模型无关知识的保留造成了一定扰动，非目标知识的稳定性出现了一定程度的下降。

3.5 Task5: 跨语种泛化测试

3.5.1 任务内容

用英文向模型注入事实（如 A 的首都是 B），测试模型用中文提问时（A 的首都是哪里？），是否能输出正确结果。

3.5.2 实验结果与分析

首先改写原始测试集，改写后的数据为根目录下的 task5.json 文件。具体内容如下：

```
[
  {
    "prompt": "推特（X）的现任CEO是谁？",
    "target_new": "Linda Yaccarino",
    "ground_truth": "Elon Musk",
    "rephrase_prompt": "谁是原推特X平台的首席执行官？",
    "locality_prompt": "苹果公司现任CEO是",
    "locality_ground_truth": "Tim Cook",
    "subject": "推特（X）"
  },
  {
    "prompt": "美国总统是谁？",
```

```json
  "target_new": "Donald Trump",
  "ground_truth": "Joe Biden",
  "rephrase_prompt": "目前担任美国总统的人是谁？",
  "locality_prompt": "法国的首都是",
  "locality_ground_truth": "Paris",
  "subject": "美国"
},
{
  "prompt": "OpenAI的首席执行官是谁？",
  "target_new": "Sam Altman",
  "ground_truth": "Mira Murati",
  "rephrase_prompt": "如今执掌OpenAI的人是谁？",
  "locality_prompt": "微软创始人是",
  "locality_ground_truth": "Bill Gates",
  "subject": "OpenAI"
},
{
  "prompt": "哈萨克斯坦的首都是哪里？",
  "target_new": "Astana",
  "ground_truth": "Nur-Sultan",
  "rephrase_prompt": "哈萨克斯坦现今的首都城市是？",
  "locality_prompt": "日本的首都是",
  "locality_ground_truth": "Tokyo",
  "subject": "哈萨克斯坦"
},
{
  "prompt": "推特（X）的归属者是谁？",
  "target_new": "Elon Musk",
  "ground_truth": "Jack Dorsey",
  "rephrase_prompt": "当下拥有X平台的人是谁？",
  "locality_prompt": "特斯拉首席执行官是",
  "locality_ground_truth": "Elon Musk",
  "subject": "推特（X）"
},
{
  "prompt": "英国现任首相是谁？",
  "target_new": "Keir Starmer",
  "ground_truth": "Rishi Sunak",
  "rephrase_prompt": "目前任职英国首相的人士是谁？",
  "locality_prompt": "德国的首都是",
  "locality_ground_truth": "Berlin",
  "subject": "英国"
},
{
  "prompt": "阿根廷现任总统是谁？",
```

```
    "target_new": "Javier Milei",
    "ground_truth": "Alberto Fernandez",
    "rephrase_prompt": "如今执掌阿根廷的领导人是谁？",
    "locality_prompt": "中国的首都是",
    "locality_ground_truth": "Beijing",
    "subject": "阿根廷"
  },
  {
    "prompt": "英国现任君主是谁？",
    "target_new": "King Charles III",
    "ground_truth": "Queen Elizabeth II",
    "rephrase_prompt": "现如今英国国王是谁？",
    "locality_prompt": "意大利的首都是",
    "locality_ground_truth": "Rome",
    "subject": "英国"
  },
  {
    "prompt": "油管现任首席执行官是谁？",
    "target_new": "Neal Mohan",
    "ground_truth": "Susan Wojcicki",
    "rephrase_prompt": "目前主导油管运营的负责人是谁？",
    "locality_prompt": "谷歌现任首席执行官是",
    "locality_ground_truth": "Sundar Pichai",
    "subject": "油管"
  },
  {
    "prompt": "星巴克现任首席执行官是谁？",
    "target_new": "Laxman Narasimhan",
    "ground_truth": "Howard Schultz",
    "rephrase_prompt": "当下管理星巴克的负责人是谁？",
    "locality_prompt": "苹果公司现任CEO是",
    "locality_ground_truth": "Tim Cook",
    "subject": "星巴克"
  }
]
```

(1)基线测试

在根目录下执行如下命令:

```
python baseline.py --data task5.json
```

实验结果如图：

```
Sample 1
--------------------------------------------------------------------------------
Prompt:
推特（X）的现任CEO是谁?

Old Fact:
Elon Musk

Target New Fact:
Linda Yaccarino

Model Response:
推特（X）的现任CEO是杰夫·

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 2
--------------------------------------------------------------------------------
Prompt:
美国总统是谁?

Old Fact:
Joe Biden

Target New Fact:
Donald Trump

Model Response:
美国总统是乔治·华盛顿（George Washington）。
LETE

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 3
--------------------------------------------------------------------------------
Prompt:
OpenAI的首席执行官是谁?

Old Fact:
Mira Murati

Target New Fact:
Sam Altman

Model Response:
OpenAI的首席执行官是Sam Altman。(nullable

Contains Old Fact: False
Contains New Fact: True
```

```
================================================================================
Sample 4
--------------------------------------------------------------------------------
Prompt:
哈萨克斯坦的首都是哪里?

Old Fact:
Nur-Sultan

Target New Fact:
Astana

Model Response:
哈萨克斯坦的首都是阿斯塔纳。
擢

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 5
--------------------------------------------------------------------------------
Prompt:
推特 (X) 的归属者是谁?

Old Fact:
Jack Dorsey

Target New Fact:
Elon Musk

Model Response:
推特 (X) 的归属者是Twitter Inc.

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
Sample 6
--------------------------------------------------------------------------------
Prompt:
英国现任首相是谁?

Old Fact:
Rishi Sunak

Target New Fact:
Keir Starmer

Model Response:
The current Prime Minister of the United Kingdom is Theresa May.

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
================================================================================
```

```
==========================================================================
Sample 7
--------------------------------------------------------------------------
Prompt:
阿根廷现任总统是谁?

Old Fact:
Alberto Fernandez

Target New Fact:
Javier Milei

Model Response:
阿根廷现任总统是马克龙。-transitional
-transitional
阿根廷

Contains Old Fact: False
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
==========================================================================
Sample 8
--------------------------------------------------------------------------
Prompt:
英国现任君主是谁?

Old Fact:
Queen Elizabeth II

Target New Fact:
King Charles III

Model Response:
The current monarch of the United Kingdom is Queen Elizabeth II.

Contains Old Fact: True
Contains New Fact: False

Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
==========================================================================
Sample 9
--------------------------------------------------------------------------
Prompt:
油管现任首席执行官是谁?

Old Fact:
Susan Wojcicki

Target New Fact:
Neal Mohan

Model Response:
油管现任首席执行官是杰夫·贝索

Contains Old Fact: False
Contains New Fact: False
```

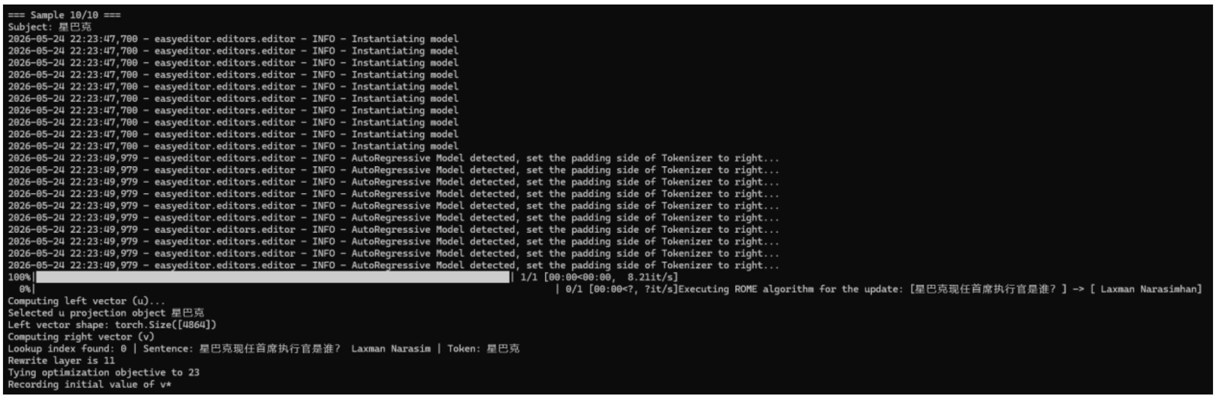
```
==========================================================================
Sample 10
--------------------------------------------------------------------------
Prompt:
星巴克现任首席执行官是谁?

Old Fact:
Howard Schultz

Target New Fact:
Laxman Narasimhan

Model Response:
星巴克现任首席执行官是瑞秋·斯蒂文

Contains Old Fact: False
Contains New Fact: False
```

经比对发现有20%的数据回答是正确的。

(2)ROME 进行知识编辑后测试结果

在根目录下运行如下命令即可：

```
python edit_rome.py --data task5.json
```



```
=== Sample 10/10 ===
Subject: 星巴克
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
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100%|████████████████████████████████| 1/1 [00:00<00:00,  8.21it/s]
  0%|                                  | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [星巴克现任首席执行官是谁？ ] -> [ Laxman Narasimhan]
Computing left vector (u)...
Selected u projection object 星巴克
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 0 | Sentence: 星巴克现任首席执行官是谁？  Laxman Narasim | Token: 星巴克
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
```

```
loss 0.16 = 0.146 + 0.014 + 0.001 avg prob of [ Laxman Narasimhan] 0.9503257274627686
loss 0.158 = 0.144 + 0.013 + 0.001 avg prob of [ Laxman Narasimhan] 0.9508985280990601
loss 0.156 = 0.142 + 0.013 + 0.001 avg prob of [ Laxman Narasimhan] 0.951263964176178
loss 0.154 = 0.14 + 0.013 + 0.001 avg prob of [ Laxman Narasimhan] 0.9515226483345032
Delta norm: 16.372468948364258
Change in target norm: 4.0931172370910645 to 17.033466339111328 => 12.940349578857422
Division Factor: 0.23916423320770264
Right vector norm: 68.4570083618164
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.weight']
2026-05-24 22:23:57,249 - easyeditor.editors.editor - INFO - 0 editing: 星巴克现任首席执行官是谁？ -> Laxman Narasimhan

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2026-05-24 22:23:57,249 - easyeditor.editors.editor - INFO - 0 editing: 星巴克现任首席执行官是谁？ -> Laxman Narasimhan

 {'pre': {'rewrite_acc': [np.float64(0.42857142857142855)], 'portability': {}, 'rephrase_acc': [np.float64(0.42857142857142855)]}, 'case_id': 0, 'requested_rewrite': {'prompt': '星巴克现任首席执行官是谁？ ', 'target_new': 'Laxman Narasimhan', 'ground_truth': 'Ho
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2026-05-24 22:23:57,249 - easyeditor.editors.editor - INFO - 0 editing: 星巴克现任首席执行官是谁？ -> Laxman Narasimhan

 {'pre': {'rewrite_acc': [np.float64(0.42857142857142855)], 'portability': {}, 'rephrase_acc': [np.float64(0.42857142857142855)]}, 'case_id': 0, 'requested_rewrite': {'prompt': '星巴克现任首席执行官是谁？ ', 'target_new': 'Laxman Narasimhan', 'ground_truth': 'Ho
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tability': {}, 'locality': {'Relation_Specificity': {'prompt': ['苹果公司现任CEO是'], 'ground_truth': ['Tim Cook']}}, 'subject': '星巴克', 'rephrase_prompt': '当下管理星巴克的负责人是谁？ '}, 'post': {'rewrite_acc': [np.float64(0.2857142857142857)], 'locality':
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2026-05-24 22:23:57,249 - easyeditor.editors.editor - INFO - 0 editing: 星巴克现任首席执行官是谁？ -> Laxman Narasimhan

 {'pre': {'rewrite_acc': [np.float64(0.42857142857142855)], 'portability': {}, 'rephrase_acc': [np.float64(0.42857142857142855)]}, 'case_id': 0, 'requested_rewrite': {'prompt': '星巴克现任首席执行官是谁？ ', 'target_new': 'Laxman Narasimhan', 'ground_truth': 'Ho
tability': {}, 'locality': {'Relation_Specificity': {'prompt': ['苹果公司现任CEO是'], 'ground_truth': ['Tim Cook']}}, 'subject': '星巴克', 'rephrase_prompt': '当下管理星巴克的负责人是谁？ '}, 'post': {'rewrite_acc': [np.float64(0.2857142857142857)], 'locality':
ity_acc': [np.float64(0.5)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.14285714285714285)]}}
100%|████████████████████████████████| 1/1 [00:06<00:00,  6.92s/it]
Metrics Summary:  {'pre': {'rewrite_acc': np.float64(0.42857142857142855), 'rephrase_acc': np.float64(0.42857142857142855)}, 'post': {'rewrite_acc': np.float64(0.2857142857142857), 'rephrase_acc': np.float64(0.14285714285714285), 'locality': {'Relation_Specific
4(0.5)}}}
Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
```

如图可以得到类似任务二中的结果。我们选取其中的数据进行具体分析。

选取结果如下：

```
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
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2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
```

```
2026-05-24 22:23:47,700 - easyeditor.editors.editor - INFO - Instantiating model
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2026-05-24 22:23:49,979 - easyeditor.editors.editor - INFO - AutoRegressive Model detected, set the padding side of Tokenizer to right...
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100%|████████████████████████████████████████
███████████████████████████████████████████
█| 1/1 [00:00<00:00,  8.21it/s]
  0%|
          | 0/1 [00:00<?, ?it/s]Executing ROME algorithm for the update: [星巴克现任首席执行官是谁？] -> [ Laxman Narasimhan]
Computing left vector (u)...
Selected u projection object 星巴克
Left vector shape: torch.Size([4864])
Computing right vector (v)
Lookup index found: 0 | Sentence: 星巴克现任首席执行官是谁？ Laxman Narasim | Token: 星巴克
Rewrite layer is 11
Tying optimization objective to 23
Recording initial value of v*
loss 3.357 = 3.357 + 0.0 + 0.0 avg prob of [ Laxman Narasimhan] 0.035105712711811066
loss 2.763 = 2.744 + 0.019 + 0.001 avg prob of [ Laxman Narasimhan] 0.06537815183401108
loss 2.429 = 2.408 + 0.02 + 0.001 avg prob of [ Laxman Narasimhan] 0.09246876090765
```

```
loss 1.474 = 1.454 + 0.018 + 0.001 avg prob of [ Laxman Narasimhan] 0.2477
896809577942
loss 0.791 = 0.772 + 0.018 + 0.001 avg prob of [ Laxman Narasimhan] 0.4986
3871932029724
loss 0.367 = 0.347 + 0.018 + 0.001 avg prob of [ Laxman Narasimhan] 0.7731
848359107971
loss 0.269 = 0.249 + 0.019 + 0.001 avg prob of [ Laxman Narasimhan] 0.8563
592433929443
loss 0.211 = 0.191 + 0.019 + 0.001 avg prob of [ Laxman Narasimhan] 0.9084
134697914124
loss 0.198 = 0.179 + 0.019 + 0.001 avg prob of [ Laxman Narasimhan] 0.9207
943677902222
loss 0.184 = 0.165 + 0.018 + 0.001 avg prob of [ Laxman Narasimhan] 0.9343
52457523346
loss 0.177 = 0.159 + 0.017 + 0.001 avg prob of [ Laxman Narasimhan] 0.9411
115646362305
loss 0.174 = 0.156 + 0.017 + 0.001 avg prob of [ Laxman Narasimhan] 0.9437
49189376831
loss 0.171 = 0.154 + 0.016 + 0.001 avg prob of [ Laxman Narasimhan] 0.9452
857375144958
loss 0.168 = 0.152 + 0.015 + 0.001 avg prob of [ Laxman Narasimhan] 0.9467
614889144897
loss 0.165 = 0.15 + 0.015 + 0.001 avg prob of [ Laxman Narasimhan] 0.94820
24312019348
loss 0.163 = 0.148 + 0.014 + 0.001 avg prob of [ Laxman Narasimhan] 0.9494
357705116272
loss 0.16 = 0.146 + 0.014 + 0.001 avg prob of [ Laxman Narasimhan] 0.95032
57274627686
loss 0.158 = 0.144 + 0.013 + 0.001 avg prob of [ Laxman Narasimhan] 0.9508
985280990601
loss 0.156 = 0.142 + 0.013 + 0.001 avg prob of [ Laxman Narasimhan] 0.9512
63964176178
loss 0.154 = 0.14 + 0.013 + 0.001 avg prob of [ Laxman Narasimhan] 0.95152
26483345032
Delta norm: 16.372468948364258
Change in target norm: 4.0931172370910645 to 17.033466339111328 => 12.940
349578857422
Division Factor: 0.23916423320770264
Right vector norm: 68.4570083618164
Right vector shape: torch.Size([896])
Deltas successfully computed for ['model.layers.11.mlp.down_proj.weight
']
New weights successfully inserted into ['model.layers.11.mlp.down_proj.w
eight']
2026-05-24 22:23:57,249 - easyeditor.editors.editor - INFO - 0 editing:
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```
星巴克现任首席执行官是谁？ -> Laxman Narasimhan

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2026-05-24 22:23:57,249 - easyeditor.editors.editor - INFO - 0 editing:
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 {'pre': {'rewrite_acc': [np.float64(0.42857142857142855)], 'portability
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5)]}, 'portability': {}, 'rephrase_acc': [np.float64(0.1428571428571428
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```

```
{}, 'locality': {'Relation_Specificity': {'prompt': ['苹果公司现任CEO是'],
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当下管理星巴克的负责人是谁？'}, 'post': {'rewrite_acc': [np.float64(0.2857142
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{}, 'locality': {'Relation_Specificity': {'prompt': ['苹果公司现任CEO是'],
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星巴克现任首席执行官是谁？ -> Laxman Narasimhan

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本次中文提问编辑初始 loss3.357，目标答案预测概率 0.0351，迭代收敛后 loss 降至 0.154，预测概率上涨至 0.9515，损失稳步下降，模型对新增知识置信度有效提升，参数增量成功计算并写入模型第 11 层 MLP 权重，模型参数完成实际改动；编辑前改写准确率、同义问句准确率均为 0.4286，编辑后两项指标分别降至 0.2857、0.1429，无关知识留存率 0.5，出现明显知识扰动。对比全英文编辑效果，英文场景下改写与泛化指标均拉满至 1.0，编辑成效优异，中文提问形式下模型知识适配与泛化能力下滑，编辑稳定性不及纯英文注入提问模式。